Timing, Adherence, and Hormone Profiles in a Novel, At-Home Saliva Collection Device

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Introduction

The investigator provided data from 31 healthy control subjects who collected saliva samples at home using the Saliva Procurement and Integrated Testing (SPIT) booklet, measuring cortisol and DHEA concentrations. Participants recorded sampling times in the booklet, and an electronic monitoring cap independently recorded bottle-opening times. Samples were scheduled at wake, 30 minutes after waking (as reported in sleep diary), before lunch, and 600 minutes (10 hours) after waking. Time of waking and lunch varied by subject.

The scientific questions were: 1) How closely do the booklet reported sampling times agree with electronic cap times?, 2) Do participants adhere to the 30 minute and 10 hour sampling times required by a study protocol?, 3) How cortisol and DHEA change from wake to 30 minutes and from 30 minutes to 10 hours? These translate into three statistical tasks: 1) Testing whether booklet and cap times showed perfect agreement (intercept = 0, slope = 1) using a linear mixed model (LMM), 2) Summarizing the proportion of samples within $\pm$ 7.5 minutes and $\pm$ 15 minutes of the 30 minute and 10 hour targets, and 3) Estimating the change in cortisol and DHEA from wake to 30 minutes and from 30 minutes to 10 hours using a piecewise LMM.

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Methods

All analyses were conducted using R v4.5.1 (2025-06-13 ucrt). Data management included removal of DHEA values at the assay's upper detection limit (5.205 nmol/L) and exclusion of subjects with at least two such values due to possible underlying conditions. Cortisol values exceeding 80 nmol/L were removed due to implausibility. Reported sampling times from the booklet and electronic cap and reported wake time from the sleep diary were converted to POSIXct timestamps. For each sample, three variables were computed: 1) booklet time minus wake time, 2) cap time minus wake time, 3) booklet time minus cap time. All differences were calculated in minutes. These processed variables were used for all subsequent analyses.

Research Question 1: To evaluate the agreement between participant-reported sampling times in the booklet and reported sampling times from the electronic cap, a LMM was fitted with booklet time from wake as the outcome and cap time from wake as the primary predictor. A random intercept for subject was included to account for repeated measurements within individuals. An LMM was chosen as it accounts for within-subject correlation, unequal observations per participant, and allows for values to be missing at random. The primary goal was to test if booklet and cap times were in perfect agreement, corresponding to two null hypotheses: 1) The intercept is equal to 0, indicating no systematic bias between the two time sources., 2) The slope is equal to 1, indicating a one-to-one relationship between booklet and cap times. A significance threshold of $\alpha = 0.05$ was used.

Research Question 2: Adherence to the required 30 minute and 10 hour samples was descriptively assessed. First, the difference between the booklet and cap time for each sample was summarized to quantify reporting consistency. For each sample, the proportion of observations falling within $\pm$ 7.5 minutes and within $\pm$ 15 min at the second and fourth collection sample was calculated, which were intended to correspond to the 30 minute and 10 hour measurements. Second, each recorded time was separately compared to the targeted sampling time relative to the reported wake time in the sleep diary. For each target, the proportion of observations falling within $\pm$ 7.5 minutes and within $\pm$ 15 min was computed. These summaries directly quantify participant compliance and, thus, the feasibility of the sampling protocol, all without requiring parametric modeling assumptions.

Research Question 3: To characterize changes in cortisol and DHEA across the day, piecewise LMMs were fitted with a knot at 30 minutes post-wake as reported by the sleep diary. The 30 minute knot was chosen to correspond to the expected patterns of cortisol and DHEA over the course of a day. The primary predictor was booklet time from wake, chosen as it had fewer missing values than cap time and aligned more closely with the sleep diary. Two time variables were constructed, representing minutes from wake to 30 minutes and minutes beyond 30 minutes. The distinct time variables allowed for the estimation of separate linear slopes for the wake to 30 minute and after 30 minute measurements across the day, accounting for the hormones' diurnal nature. Cortisol and DHEA concentrations were log transformed prior to modeling to improve normality. Each hormone was modeled with the log concentration as the outcome and a random intercept for subject to account for repeated measures.

Results

Table 1 summarizes key variables and missingness across the four collection times. Missingness was highest for the cap time measurements, particularly in samples 1 and 2. Hormone concentrations had the expected decline across the day, with highest cortisol and DHEA values in samples 1 and 2. One participant (ID 3037) was excluded due to four DHEA measurements at the assay's upper detection limit.

Research Question 1: A LMM was used to assess agreement between the recorded booklet and cap sampling times. The slope differed significantly from 0 ($p < 0.001$). For every one minute increase in cap time, booklet time increased by 0.994 minutes, on average (95% CI: [0.98, 1.01]). As the 95% CI includes 1, this is a near one-to-one relationship. However, the intercept differed significantly from 0 ($p = 0.028$), indicating that booklet times were, on average, recorded 6.7 minutes earlier than cap times at wake (95/% CI: [-12.53, -0.88]). *Figure 1* illustrates this systematic bias and one-to-one relationship.

Research Question 2: When examining the difference between booklet and cap times for the second and fourth sample of the day (intended to occur 30 minutes and 10 hours post wake), 59-64% of samples fell within $\pm$ 7.5 minutes and 75-80% fell within $\pm$ 15 minutes. These values indicate moderate agreement between the two time sources, consistent with the findings from *Research Question 1*.

Adherence to the intended sampling times was evaluated separately for booklet and cap times relative to the reported wake time (*Figure 2*). Booklet times consistently demonstrated higher adherence than cap times. At the 30 minute sample, 78.6% of booklet times fell within ± 7.5 minutes of the target compared to 52.9% of cap times. For ± 15 minutes, adherence expanded to 90.5% and 70.6%, respectively. A similar pattern was observed for the 10 hour sample, though adherence was lower overall: 48.1% vs. 33.8% for the booklet and cap within ± 7.5 minutes and 55.8% vs. 41.2% within ± 15 minutes.

Research Question 3: Piecewise LMMs with a knot at 30 minutes were used to characterize diurnal trajectories of cortisol and DHEA. Both outcomes were modeled on the log scale due to right skewed distributions. Booklet times were selected as the primary time variable as they showed higher adherence and lower missingness than cap times.

For cortisol, the intercept differed significantly from 0 ($p < 0.001$), corresponding to an estimated waking concentration of 5.39 nmol/L (95% CI: [4.41, 6.58]). From wake to 30 minutes, cortisol increased by 0.62% per minute on average (95% CI: [-0.10, 1.34]), but this slope was not statistically significant ($p = 0.094$). After 30 minutes, cortisol declined significantly ($p < 0.001$) with a 12.53% decrease per hour on average (95% CI: [-14.41, -10.61]). *Figure 3* displays the fitted trajectory.

For DHEA, the intercept was not significantly different from 0 ($p = 0.067$), corresponding to an estimated DHEA concentration at wake of 1.24 nmol/L (95% CI: [0.99, 1.56]). From wake to 30 minutes, DHEA declined sharply by 1.76% per minute on average (95% CI: [-2.35, -1.17]; $p < 0.001$). After 30 minutes, DHEA continued to decline at an estimated 8.82% decrease in DHEA per hour (95% CI: [-10.45, -7.16]; $p < 0.001$). The fitted trajectory is visualized in *Figure 4*.

Discussion

This study evaluated the feasibility and accuracy of the SPIT booklet for at-home saliva collection by examining agreement between participant recorded and electronically recorded sampling times, adherence to sampling windows, and diurnal cortisol and DHEA patterns. The results indicate that booklet and cap times were closely aligned but not interchangeable, with booklet times being systematically earlier with higher adherence and lower missingness. These findings support the use of booklet times as the primary timing source for modeling hormone trajectories.

Cortisol and DHEA both declined after 30 minutes as expected, but neither showed the anticipated early increase. This pattern may reflect timing inaccuracies, participant behavior during at-home collection, or limitations of the electronic cap.

Additionally, the study lacked a definitive measure of sampling time, making it difficult to determine whether discrepancies reflected reporting error, device malfunction, or participant behavior. The sample size may limit generalizability and reduce power to detect subtle effects. Assay constraints required censoring of DHEA values at the upper detection limit, and one participant was excluded for repeated censoring. The electronic cap showed inconsistent performance, contributing to missingness and limiting its utility as a reference standard. Despite these limitations, the findings support the feasibility of the SPIT booklet and provide guidance for improving timing accuracy in future at-home endocrine studies.

Table 1: **Summary of Missingness and Key Variables by Collection Sample**

Variable	**1** N = 90[1]	**2** N = 90[1]	**3** N = 90[1]	**4** N = 90[1]
Cap – Wake Time (min)	5.11 (9.36)	50.38 (35.44)	326.46 (79.31)	659.76 (99.30)
Missing	16	22	12	10
Booklet – Wake Time (min)	0.86 (3.73)	34.29 (9.61)	325.63 (85.00)	641.01 (84.46)
Missing	5	6	11	13
Booklet – Cap Time (min)	-4.03 (8.92)	-16.56 (36.31)	-6.68 (43.48)	-6.46 (29.90)
Missing	21	26	19	20
Cortisol (nmol/L)	5.34 [3.84, 8.25]	8.91 [5.93, 11.39]	2.86 [1.43, 4.21]	1.79 [0.97, 3.26]
Missing	0	1	2	3
DHEA (nmol/L)	1.49 [0.77, 2.18]	0.93 [0.53, 1.28]	0.40 [0.19, 0.57]	0.31 [0.16, 0.48]
Missing	2	1	2	2

[1] Mean (SD); Median [Q1, Q3]

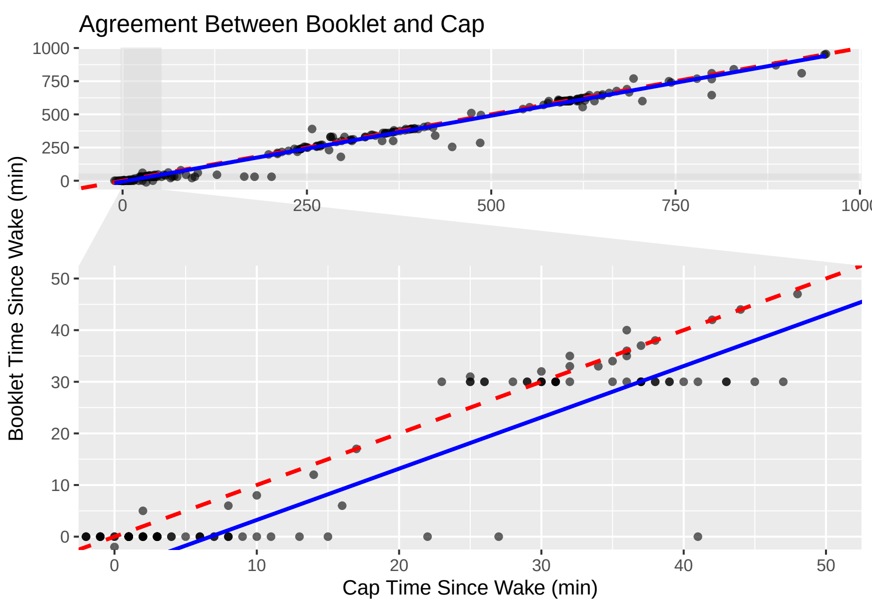

Figure 1: Research Question 1 – Agreement between reported booklet time since wake and reported cap time since wake.

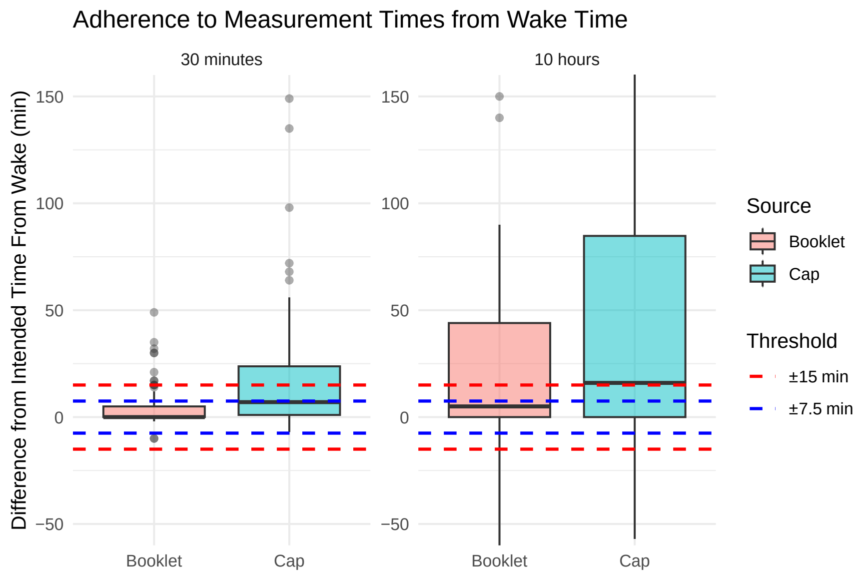

Figure 2: Research Question 2 – Adherence to 30 minute and 10 hour sampling times by booklet and cap.

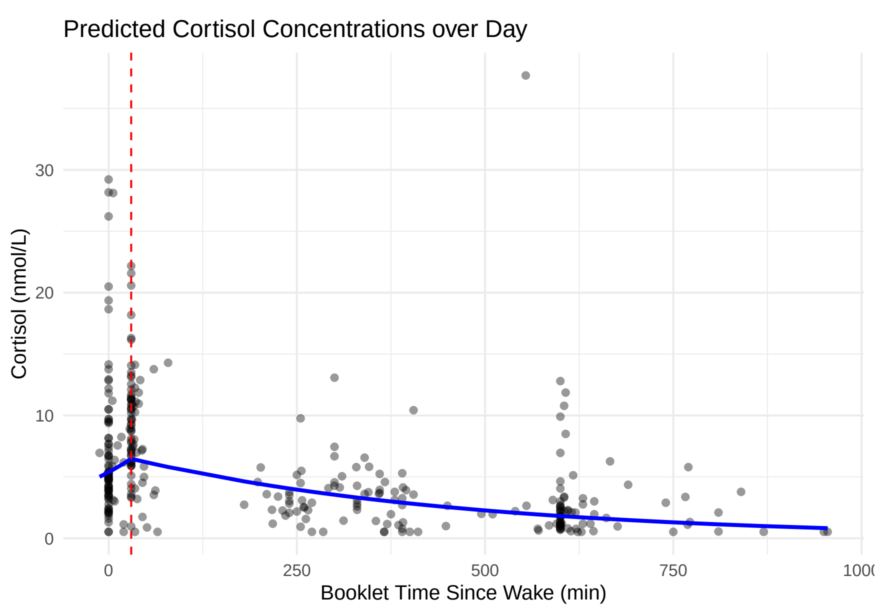

Figure 3: Research Question 3 – Predicted trend of cortisol over time since wake, reported from booklet.

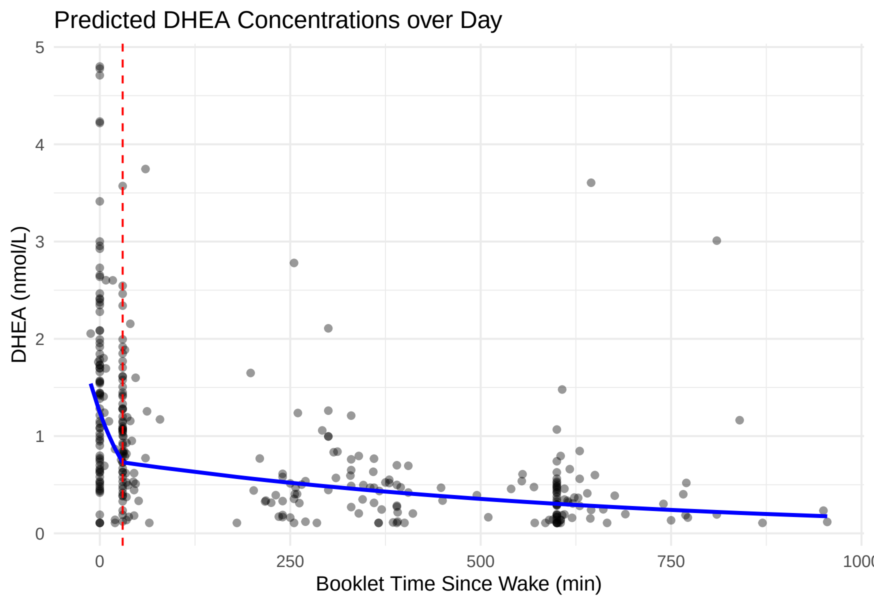

Figure 4: Research Question 3 – Predicted trend of DHEA over time since wake, reported from booklet.

Appendix

Directory: https://github.com/eweible/BIOS6624.git

File name: P0/Code/P0_Report.Rmd